Case 6: How Fast Should a Delivery Driver Drive?

Cruising Speed Policy for Regional Fleets Under Energy vs. Time Trade-Offs

Kutaisi International University | Industry Case Studies: Solving Real Problems with Mathematics

Polya Steps 1–4 Industrial Problem Canvas

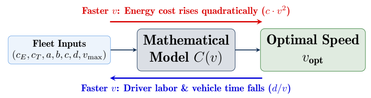
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October 2026 | Prof. Dr. Florian Rupp

Industrial Situation: The Operational Decision

A regional delivery company wants drivers to finish long motorway sections quickly. Driving faster saves paid driver time but uses more fuel or electricity, and slower driving does the opposite. The operations manager must give drivers one recommended cruising speed.

The Decision Stated Precisely: *Which cruising speed v should the company recommend on long motorway sections, and under which conditions (energy price, urgency, vehicle, route length) does this recommendation change?*



Step A: Understand (Industrial Problem Canvas)

Mapped systematically onto the lecture's Problem Canvas fields (Slide 51):

- **A.1 Situation:** Fleet operations dispatching vehicles across long, uniform motorway sections.
- **A.2 Problem:** Operational conflict between saving productive driver hours and paying for aerodynamic energy waste.
- **A.3 Decision:** Determine optimal cruising speed v and quantify sensitivity across operational parameters.
- **A.4 Stakeholders:** Operations manager (decision maker), drivers (workload, safety), finance (energy expenses), customers (delivery arrival), maintenance (wear), sustainability officer, regulator (speed limits).
- **A.5 Success (Objective):** Minimize total operational route cost $C(v)$ while strictly meeting delivery deadlines.
- **A.6 Known Information:** Transit distance $d = 300$ km, functional form of vehicle energy $E(v)$, current energy prices.
- **A.7 Unknown Information:** Physical vehicle parameters (a, b, c), true value of driver time c_T, wear and liability costs.
- **A.8 Assumptions:** Constant speed, flat motorway, calm weather, unpriced wear absorbed into maximum legal threshold $v_{\max}$.
- **A.9 Hypothesis H1:** Minimizing energy alone is not sensible ($v_E \approx 42$ km/h is far below usable motorway speeds).
- **A.9 Hypothesis H2:** At realistic values of time ($c_T > 6.3$ EUR/h), time cost dominates, pushing optimal speed to $v_{\max} = 110$ km/h.
- **A.9 Hypothesis H3:** An interior optimum exists only below a critical μ^*, and is insensitive to parameter errors (cube-root law).
- **A.9 Hypothesis H4:** Longer routes need a lower optimal speed (tested & refuted in Step 4).
- **A.10 Variables:** Decision: cruising speed v. Observables: trip time $T(v)$, energy $E(v)$. Uncertainties: a, b, c, c_E, c_T.

Warning: The wage is not the value of time.
Saved driver time only possesses true economic value if the delivery schedule can productively utilize the saved minutes.

Operational Parameters & Model Simplifications

Symbol	Physical Unit	Benchmark (Range)	Physical Meaning / Source Authority
v	km/h	70–130	Decision variable; set by driver / fleet policy
d	km	300 (150–500)	Route motorway transit distance (Dispatcher)
c_E	EUR/kWh	0.25 (0.18–0.38)	Electricity contract tariff (Commercial energy market)
c_T	EUR/h	20.0 (5–35)	Opportunity value of driver time (Wage + Urgency)
a, b, c	[phys. units]	30, 450, $3.0 \cdot 10^{-3}$	Aeromechanical drag and rolling coefficients (OEM)
$v_{\max}, T_{\max}$	km/h, h	110 km/h, 3.0 h	Highway speed limit code & delivery deadline (SLA)

Evaluation of Initial Model Simplifications (Canvas B.1):

- **Constant Speed:** Realistic on open motorway corridors; transient ramp and queue accelerations are negligible over long 300 km transits.
- **Flat Corridor:** Standard macroscopic fleet assumption; rolling elevation grades cancel out symmetrically across round-trips via regenerative braking.
- **Calm Atmospheric Conditions:** Neglects stochastic headwind/tailwind velocities; headwind w shifts effective drag to $c(v + w)^2$, slightly lowering v^*.
- **Linear Time Cost:** Time value enters linearly ($c_T d/v$); step-penalty surcharges for missed deadlines are enforced via $T(v) \le T_{\max}$.
- **Fleet Homogeneity:** Uniform vehicle coefficients; mixed fleets require calibrated (a_i, b_i, c_i) per vehicle class.

Step B: Plan (Industrial Problem Canvas)

Mapped systematically onto the lecture's Problem Canvas fields (Slide 53):

- **B.1 Simplification:** Focus on steady cruising; isolate rolling resistance (a), auxiliary loads (b/v), and aerodynamic drag ($c\,v^2$). Initially ignore acceleration transients.
- **B.2 First Model:** 1D continuous convex cost: $C(v) = c_E(a + b/v + c\,v^2) + c_T(d/v)$, with travel time $T(v) = d/v$.
- **B.3 Mathematical Toolbox:** Calculus for optimum ($C'(v) = 0, C''(v) > 0$), boundary checking ($v \le v_{\max}$), sensitivity elasticity $\frac{\mu}{v^*}\frac{dv^*}{d\mu}$. *Do not select a tool simply because we know it* (Slide 32).
- **B.4 Data (Needed vs. Have):** Have: physical vehicle model structure and benchmark parameters. Need: calibrated fleet consumption data before rollout.
- **B.5 Validation:** Verify limiting case ($\lim_{\mu \to 0} v^* = v_E$), ensure strict global convexity ($C''(v) > 0$), check feasibility against motorway speed limits.
- **B.6 Project Road-map:** Work packages: Model formulation → Analytical optimization → Constraint enforcement → Scenario evaluation → Actionable policy.
- **B.7 Time & Effort Allocation:** Prioritize economic interpretation and management decision rules over unnecessary simulation.
- **B.8 Stop Criterion:** Closed-form analytical formula for v^* and critical threshold μ^*, answering the operations manager's question across all operating regimes.

Step C: Do (Implementation & Calculations)

The Exactly Five Governing Equations (Implementation C.1):

$$C(v) = c_E\left(a + \frac{b}{v} + c\,v^2\right) + c_T\frac{d}{v} \tag{1}$$

$$v^* = \left[\frac{c_E b + c_T d}{2 c_E c}\right]^{1/3} = \left[\frac{b + \mu d}{2c}\right]^{1/3} \tag{2}$$

$$C(v^* + \delta) - C(v^*) \approx 3\,c_E c\,\delta^2 \tag{3}$$

$$\frac{\mu}{v^*}\frac{dv^*}{d\mu} = \frac{1}{3}\frac{\mu d}{b + \mu d} \le \frac{1}{3} \tag{4}$$

$$\mu^* = \frac{2c\,v_{\max}^3 - b}{d} \implies v_{\text{opt}} = \begin{cases} v^* & \mu < \mu^* \\ v_{\max} & \mu \ge \mu^* \end{cases} \tag{5}$$

Documented Benchmark Parameters (C.2, $d = 300$ km, Electric Van):
$a = 30$ kWh, $b = 450$ kWh·km/h, $c = 3.0 \cdot 10^{-3}$ kWh·h^2/km^2, $c_E = 0.25$ EUR/kWh, $v_{\max} = 110$ km/h, $T_{\max} = 3$ h.
Plausibility Checks & Comparisons (C.3 & C.4): Energy-minimal baseline $v_E = (b/(2c))^{1/3} \approx 42.2$ km/h (confirms H1). Critical operating threshold $\mu^* = (2c v_{\max}^3 - b)/d \approx 25.1$ kW $\implies c_T^* \approx 6.27$ EUR/h.

Speed	Scenario 1: $c_T = 20$ EUR/h (Standard)	Scenario 2: $c_T = 5$ EUR/h (Low Urgency)
70 km/h	98.50 EUR (+37%)	34.21 EUR (+10%)
90 km/h	81.49 EUR (+13%)	31.49 EUR (+1.2%)
100 km/h	76.13 EUR (+5.5%)	31.13 EUR (+0.05%)
110 km/h	**72.14 EUR (Best, limit binds)**	31.23 EUR (+0.4%)
$v^* = 102.4$ km/h	N/A ($v^* = 159.7$, clamped at 110)	**31.11 EUR (Best interior)**

Figure 1: Cost Trade-Off S1 vs. S2

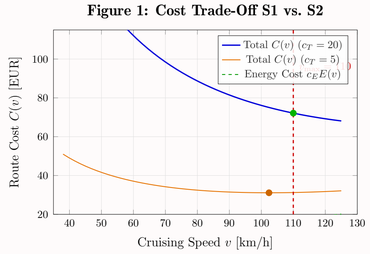
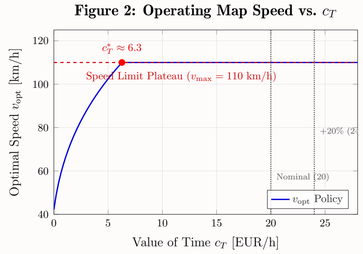


Figure 2: Operating Map Speed vs. c_T



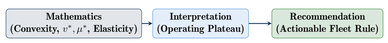
Step D: Reflect (Industrial Problem Canvas)

Mapped systematically onto the lecture's Problem Canvas fields (Slide 55):

- **D.1 Mathematical Verification:** First derivative yields closed-form root $v^* = [(b + \mu d)/(2c)]^{1/3}$. Second derivative $C''(v^*) = 6c_E c > 0$ strictly guarantees global uniqueness. Limiting case $\lim_{\mu \to 0} v^* = v_E \approx 42.2$ km/h.
- **D.2 Real-World Validation (Hypotheses):**
  - *H1 (Energy Minimum):* Supported. $v_E \approx 42$ km/h is dangerously slow and legally prohibited on motorways.
  - *H2 (Wage Dominance):* Supported. Commercial wages (15–25 EUR/h > 6.3 EUR/h) place operations on the $v_{\max}$ limit plateau.
  - *H3 (Flat Curvature):* Supported. Interior optimum exists only for low time-value ($\mu < \mu^*$). Flat curvature ($3c_E c\delta^2$) renders costs highly forgiving to small speed deviations (± 5 km/h $\implies < 0.1\%$ cost increase).
  - *H4 (Route Length):* Refuted. Route length d scales energy and time proportionally, leaving optimal cruising speed v^* independent of d.
- **D.3 Robustness:** Cost is flat near v^*. Parameter sensitivity: drag $c + 20\%$ shifts v^* down by only -5.9% (factor $1.2^{-1/3}$). Time cost $c_T + 20\%$ shifts v^* by at most $+6.3\%$.
- **D.4 Limitations:** Unpriced battery wear and traffic risk (handled via $v_{\max}$ ceiling), constant gradient, no traffic congestion shocks.
- **D.5 Interpretation:** Answers Slide 49's core question: *What changed? What matters? What should the fleet manager do on Monday morning?*
- **D.7 Confidence:** High confidence for motorway cruising; speed limit $v_{\max}$ provides a safe operational upper bound.
- **D.8 Next Steps:** Calibrate coefficients a, b, c from physical vehicle test runs before full fleet rollout.

Warning: Optimal speed ≠ Fastest speed ≠ Most energy-efficient speed.
True cost optimality balances energy expenditure against productive labor utility.

Warning: Better fit ≠ Better model (Slides 36–45).
High-degree regression without aeromechanical basis produces unphysical non-convexities.



D.6 Recommendation (Executive Policy for Monday Morning)

Official Recommendation to Operations Management (Slide 49):

"Drive at the legal limit (110 km/h) on long motorway sections whenever a saved driver-hour is worth more than about 6 EUR; at 20 EUR/h, driving 90 km/h instead raises route cost by about 13%. If deliveries have slack, drive only as fast as the deadline requires. Slower speeds save energy but cost more in time. Replace our assumed consumption data by fleet measurements before rollout."

(Exactly 66 words answering "What should the company do on Monday morning?")

Notation, Physical Quantities & Symbols

Symbol	Unit	Physical Meaning / Role	Symbol	Unit	Physical Meaning / Role
v	km/h	Cruising decision variable	c_E	EUR/kWh	Energy electricity unit tariff
v^*	km/h	Unconstrained analytical optimum	c_T	EUR/h	Opportunity value of driver time
d	km	Transit route distance	μ	kW	Time value in energy units (c_T/c_E)
$T(v)$	h	Motorway transit time (d/v)	μ^*	kW	Critical threshold where limit binds
$E(v)$	kWh	Route energy consumption	$C(v)$	EUR	Total route operational cost

Lecture Standard Clarification: Following course conventions, E denotes total route energy consumption (distinct from activation energy in Case 1); T denotes transit travel time (distinct from reactor temperature in Case 1).